

Babak Rahimi Ardabili

📍 Charlotte, NC

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in Babak Rahimi Ardabili

🔗 Google Scholar

Summary

Ph.D. in Public Policy with interdisciplinary expertise in **Statistical Analysis, Machine Learning, and Co-Creating Responsible AI**. Proven record of **writing grants** for federal agencies such as National Science Foundation (NSF). Experience in teaching and co-teaching **Data Analysis and Statistical methods** Courses. Ability to work with **stakeholders** to contribute in research activities.

Education

- Ph.D. University of North Carolina at Charlotte**, Public Policy Analysis Jan 2021 – Dec 2025
- **Dissertation:** Co-Creating Responsible AI for Public Safety
 - **Coursework:** Introduction to Machine Learning, Applied Machine Learning, Ethics of Public Policy, Research Design in Public Policy, Complex Systems in Public Policy, Policy Process, Advanced Program Evaluation
- M.Sc. University of Tehran**, Industrial Engineering - Futures Studies Engineering Sep 2014 – Sep 2017
- **Dissertation:** Analyzing and predicting the impact of political and economic variables on Tehran stock exchange market trend
 - **Coursework:** Science and Technology Policy, Data Mining, Public Policy Making, Fundamentals of Modeling and System Dynamics, Fundamentals of Game Theory
- B.Sc. Isfahan University of Technology**, Industrial Engineering - Industrial Technology Sep 2006 – Sep 2010
- **Coursework:** Operations Research 1 and 2, Engineering Probability, Engineering Statistics, Numerical Methods, Management Information System

Teaching Experience

- University of North Carolina at Charlotte (UNCC)**, Department of Public Policy Charlotte, NC Spring 2026
- Data Science and Machine Learning for Public Policy and Social Science (Undergraduate Level - 11 Phd and Master's students). Covered statistical inference, experimental methods (A/B testing), and supervised machine learning (regression, classification, neural networks) with applied public policy datasets.
- University of North Carolina at Asheville (UNCA)**, Department of Mathematics and Statistics Asheville, NC Spring 2026
- Big Data Analytics (undergraduate Level - 26 DS, CS, and Math undergrads). Covered statistical inference, experimental design, and supervised and unsupervised learning models, emphasizing probabilistic foundations, optimization techniques, and computational implementation for real-world data analysis.
- University of North Carolina at Charlotte (UNCC)**, Department of Public Policy Charlotte, NC Fall 2026 (Scheduled)
- Studio 1: Data Management and Basic Analytics (Graduate Level - 15 PhD Students). Project-based course on data management, pre-processing, variable operationalization, and analytical interpretation, preparing students for advanced regression analysis and policy modeling.

Research and Professional Experience

- University of North Carolina at Charlotte (UNCC)**, Postdoctoral Fellow at Department of Electrical and Computer Engineering, NSF CIVIC Challenge Charlotte, NC Jan 2026 – Current

- Serve as **Project Manager** for the deployment of **AI-enabled video solutions** across **10 pilot sites** in Uptown Charlotte, NC.
- Appointed **Associate Graduate Faculty**, University of North Carolina at Charlotte, with eligibility to serve on master's theses and doctoral dissertation committees.
- Supervise **two Master's students** leading **Data Analytics and Machine Learning** research initiatives.
- Mentor **two Master's students** conducting **Community Engagement and UX Research** studies.
- Lead a team of undergraduate **application developers** at Central Piedmont Community College, overseeing system implementation and deployment.

University of North Carolina at Charlotte (UNCC), Teaching Assistant

Charlotte, NC
Sep 2024 – Dec 2025

- Collaborating with faculties to design and deliver engaging course content focused on **Statistical Methods**, and **Linear Regression**.
- Co-teaching the course.

University of North Carolina at Charlotte (UNCC), Research Assistant

Charlotte, NC
Jan 2021 – Aug 2024

- **Secured 1M USD** funding through the **NSF Civic Innovation Challenge** as a co-author and contributor.
- Focused on community engagement initiatives and the product manager of cloud-based services and mobile applications for real-time video analytics in public safety contexts. [SafeC Project](#)
- Contributing to pilot studies under the NSF-funded project "Building Safe and Secure Communities through Real-Time Edge Video Analytics," emphasizing the ethical considerations of AI deployment.

Greenweb Holding, Director of Innovation and Startup Acceleration Center

Tehran, Iran
March 2020 – Dec 2020

- Led the development of digital transformation strategies to foster innovation in technology and policy solutions.
- Facilitated workshops and mentorship programs for startups in the fields of AI and technology, promoting a culture of collaborative research and development.

Hamtech (AI Accelerator), Project Manager

Tehran, Iran
Jan 2020 – Dec 2020

- Managed project focused on developing an online accelerator SaaS platform, enhancing the integration of AI technologies in startup ecosystems.
- Conducted comprehensive due diligence for AI and data analytics startups, evaluating their technical feasibility and business viability to inform investment strategies.

IDEHHUB Investment Platform (Business Consulting Group), Senior Business Analyst Consultant

Tehran, Iran
March 2020 – Nov 2020

- Conducting Commercial Research Project "COVID-19 outlook series; effect of COVID-19 outbreak on Iranian startups".
- Conducting Commercial Research Project "Iran's Fintech ecosystem vs. global Fintech ecosystem".

Business Innovation Center of University of Tehran, Deputy Head; Due to the COVID-19 breakout the cooperation finished in 2 Months

Tehran, Iran
Jan 2020 – March 2020

- Collaborated with stakeholders to identify and develop innovative business solutions

Deputy of Policy Making and Development, General Director

Tehran, Iran
Oct 2018 – Dec 2019

- Oversaw policy development initiatives aimed at promoting science and technology innovation within the national agenda

- Engaged with policymakers and stakeholders to develop evidence-based policy recommendations.

Digital Economy and Smart Technology Development Council (DEST), Executive Manager

Tehran, Iran
Sep 2017 – Sep 2018

- Directed research initiatives to assess the economic implications of emerging technologies.
- Established collaborative partnerships with technology firms to foster a shared understanding of innovation policy.

Iran's Association for Management of Technology (IRAMOT), Quantitative Analyst

Tehran, Iran
Sep 2015 – Aug 2017

- Engaged with academic and industry experts to refine metrics for evaluating technology policy outcomes.
- Collaborated with stakeholders to disseminate findings and promote evidence-based practices in technology management.

Emtidad Engineering Company (EEC), System and Process Design Expert

Tehran, Iran
May 2013 – Aug 2015

- Designed and optimized systems and processes to enhance operational efficiency within technology projects.
- Engaged in cross-disciplinary collaboration to integrate innovative solutions in project design.

Publications

Co-Creating Responsible AI for Public Safety: A Mixed-Method Approach Toward Understanding Stakeholder Perceptions,

August 2026

Technological Forecasting and Social Change, 2026

B.R. Ardabili, A. Pazho, Jeri Guido, Philip Otienoburu, Jason Windett, Shannon Reid, H. Tabkhi

Conducted a large-scale **mixed-methods study** on stakeholder perceptions of **AI-assisted video surveillance (A²VS)**, integrating **LLM-driven thematic analysis** (LLaMA 3.1) of focus group transcripts with **random forest** of a 410-respondent public survey. Introduced the novel **Alignment-Tension-Gap (ATG) quadrant framework** to synthesize cross-stakeholder consensus, disagreement, and policy blind spots. Translated empirical findings into actionable **responsible AI governance** recommendations, advocating for a unified federal framework with national standards and civil rights protections for public safety AI deployment.

Beyond the Interface: Redefining UX for Society-in-the-Loop AI Systems,

June 2026

Proceedings of the International Conference on Sustainability, Innovation, and Society (ICSIS 2026), 2026

N. Mafi, S. Maleki, **B.R. Ardabili**, H. Tabkhi

Introduced a novel **sociotechnical UX framework** for **Human-in-the-Loop (HITL) AI systems**, combining a mixed-methods analysis of 269 stakeholder insights with the deployment of a real-world **HITL video anomaly detection pipeline**. Positioned **explainability**, **trust calibration**, and **human-AI collaboration** as first-class UX design strategies, moving beyond traditional interface-centric evaluation toward a **Society-in-the-Loop paradigm**. Formalized four novel evaluative metrics, **Accuracy (FPR/FNR)**, **Operational Latency**, **Adaptation Time**, and **Trust-in-Automation**. [Redefining UX](#)

A Case Study in Responsible AI-Assisted Video Solutions: Multi-Metric Behavioral Insights in a Public Market Setting,

June 2026

Proceedings of the International Conference on Human-AI Interaction and Experience Design (HAXD26), 2026

E. Ebong, M. Fereydouni, S. Maleki, P. Otienoburu, **B.R. Ardabili**, H. Tabkhi

Presented a real-world case study on **responsible AI deployment** for **privacy-preserving video analytics** in a city-center public market. Extracted **multi-metric behavioral insights**, including dwell time, directional flow, and **movement pattern analysis**, from abstract **pose-based representations** using geometric normalization and **trajectory clustering**. Demonstrating how **privacy-preserving computer vision** can deliver actionable operational insights for urban and retail environments. [Multi-Metric Behavioral Insights](#)

From Frames to Events: Rethinking Evaluation in Human-Centric Video Anomaly Detection,

June 2026

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026

N. Rashvand, S. Yao, A. Pazho, **B.R. Ardabili**, H. Tabkhi

Proposed an **event-centric evaluation framework** for **pose-based Video Anomaly Detection (VAD)**, addressing the misalignment between conventional **frame-level metrics** and real-world deployment needs. Introduced a score-refinement pipeline and an end-to-end **Dual-Branch reconstruction model** for **temporal event localization**, along with the first **tIoU-based event matching** and **multi-threshold F1 evaluation** protocol for VAD. Results reveal that state-of-the-art models with over 61% frame-level AUC-ROC achieve event-level precision below 10%, exposing a critical performance gap hidden by traditional evaluation. [From Frames to Events](#)

Towards Adaptive Human-centric Video Anomaly Detection: A Comprehensive Framework and A New Benchmark,

November 2025

IEEE Transactions on Circuits and Systems for Video Technology, 2025

G.A. Noghre, S. Yao, A.D. Pazho,, **B.R. Ardabili**, V. Katariya, H. Tabkhi

Planning, recording, collecting, labeling, and annotating a comprehensive **real-world anomaly detection dataset** offering a **privacy-centric approach** to AI-based anomaly detection, addressing critical concerns about **data privacy and human-centric design** in surveillance technology.[HUVAD](#)

From lab to field: Real-world evaluation of an AI-driven Smart Video Solution to enhance community safety,

September 2025

Internet of Things, 2025

S. Yao, **B.R. Ardabili**, A.D. Pazho, G.A. Noghre, C. Neff, L. Bourque, H. Tabkhi

Evaluated an **AI-enabled Smart Video Solution** that integrates with existing camera networks to enhance safety. The system prioritizes **privacy and ethics** by using pose-based data for anomaly detection. Deployed a cloud-based infrastructure and mobile app for **real-time community alerts**. Employed advanced **data visualization techniques**, including occupancy indicators, statistical anomaly detection, and heatmaps, to analyze pedestrian behaviors. Demonstrated the SVS's ability to transform **complex computer vision** outputs into actionable insights for stakeholders, law enforcement, urban planners, and social scientists.[From Lab to Field](#)

Shopformer: Transformer-Based Framework for Detecting Shoplifting via Human Pose,

July 2025

Proceedings of the Computer Vision and Pattern Recognition Conference, 2025

Narges Rashvand, G.A. Noghre, A.D. Pazho, **B.R. Ardabili**, H. Tabkhi

Developed Shopformer, a **transformer-based model** for **privacy-preserving shoplifting detection** using human pose sequences. The framework employs a **Graph Convolutional Autoencoder tokenizer** and transformer encoder-decoder to capture spatio-temporal behavior patterns, outperforming state-of-the-art anomaly detection models on the PoseLift dataset. [ShopFormer](#)

Demographic disparities, service efficiency, safety, and user satisfaction in public bus transit system: A survey-based case study in the city of Charlotte, NC,

January 2025

Transportation Research Interdisciplinary Perspectives 29, 101296

SS Hosseini **B.R. Ardabili**, M Azarbayjani, H Tabkhi

This paper explores how **public bus service** is being utilized by different **demographic groups** through a **comprehensive survey**. Leveraging statistical quantitative methods it aims to understand the gap among different demographic groups and provide **data-driven technology-oriented policy implications** to address the problem. [Public Transportation Survey](#)

Exploring Public's Perception of Safety and Video Surveillance Technology: A Survey Approach,
Technology in Society, 2024

September 2024

B.R. Ardabili, A.D. Pazho, G.A. Noghre, V. Katariya, G. Hall, S. Reid, H. Tabkhi

This paper presents the results of a **public survey** on the **societal implications** of video surveillance technologies, providing valuable data on **community** concerns, safety perceptions, and the need for equitable **AI governance**. [Public Survey](#)

Expert with Clustering: Hierarchical Online Preference Learning Framework,
6th Annual Conference on Learning for Dynamics and Control, 2024

June 2024

T Zhou, JH Cho **B.R. Ardabili**, H Tabkhi, C Wu In collaboration with Massachusetts Institute of Technology a novel **hierarchical clustering framework** for online preference learning, enhancing decision-making capabilities by dynamically categorizing user preferences is proposed. The model **integrates AI techniques to optimize recommendation systems**, offering a scalable solution for personalized, data-driven insights in various applications. [Proceedings of Machine Learning Research](#)

Understanding the Transit Gap: A Comparative Study of On-Demand Bus Services and Urban Climate Resilience in South End, Charlotte, NC, and Avondale, Chattanooga, TN,
PLEA 2024: (Re)thinking Resilience, 2024

June 2024

SS Hosseini **B.R. Ardabili**, M Azarbayjani, S Pulugurtha, H Tabkhi

Providing a **comparative analysis of on-demand bus services** in two urban areas, focusing on **transit disparities, service efficiency, and their contribution to urban climate resilience**. It highlights how smart mobility solutions can address demographic disparities and foster sustainable, resilient urban environments. [PLEA-2024](#)

Real-world community-in-the-loop smart video surveillance system,
2023 IEEE International Conference on Smart Computing (SMARTCOMP), 2023

June 2023

S Yao **B.R. Ardabili**, AD Pazho, GA Noghreh, C Neff, H Tabkhi

The study introduced a **community-in-the-loop** framework combining AI-driven behavioral analysis with a smartphone notification app, enabling **ethical, privacy-aware deployment**. The system achieved low-latency performance across multi-camera configurations and provided practical guidelines for **real-world, human-centered** surveillance implementation. [SMARTCOMP](#)

Understanding Policy and Technical Aspects of AI-Enabled Smart Video Surveillance to Address Public Safety,
Computational Urban Science, 2023

May 2023

B.R. Ardabili, A.D. Pazho, G.A. Noghre, C. Neff, S.D. Bhaskararayani, H. Tabkhi

This paper explores the **intersection of policy and technical challenges** in using AI-powered video surveillance. It provides a **framework for balancing public safety with privacy and ethical concerns**, making it highly relevant to discussions around AI regulation. [Springer.023.00097-8](#)

Chad: Charlotte Anomaly Dataset,
Scandinavian Conference on Image Analysis, 2023

April 2023

Armin Danesh Pazho, Ghazal Alinezhad Noghre **B.R. Ardabili**, Christopher Neff, H. Tabkhi

This dataset provides the real-world data required for developing and training **AI models for anomaly detection** in surveillance systems. It serves as a key resource for researchers and policymakers interested in the practical implications of AI deployments in public spaces. [CHAD](#) 

Ancilia: Scalable Intelligent Video Surveillance for the Artificial Intelligence of Things, IEEE Internet of Things Journal, 2023

March 2023

A.D. Pazho, C. Neff, G.A. Noghre, **B.R. Ardabili**, S. Yao, M. Baharani, H. Tabkhi

This work introduces a scalable **IoT surveillance framework** integrating deep learning and distributed edge–cloud computing for real-time multi-camera analysis. The system leverages **object detection (YOLOv5)**, **tracking (DeepSORT)**, **pose estimation (OpenPose)**, and **behavior anomaly detection models** within an optimized inference pipeline, achieving high **throughput (>30 FPS)** and **sub-second latency**. [Ancilia](#) 

Understanding Ethics, Privacy, and Regulations in Smart Video Surveillance for Public Safety,

December 2022

Association for the Advancement of Artificial Intelligence (AAAI), 2023

B.R. Ardabili, Armin Danesh Pazho, Ghazal Alinezhad Noghre, Christopher Neff, Arun Ravindran, H. Tabkhi

Focused on the **ethical and regulatory challenges of AI-driven surveillance**, this paper offers insights into creating **responsible AI systems that comply with privacy laws and ethical guidelines** key areas for AI policy-making. [AI for Social Good](#) 

Science, Technology, and Innovation Policy Review: Iran,

December, 2016

United Nations Conference on Trade and Development (UNCTAD), 2016

A comprehensive review commissioned by the UN, this report provides strategic recommendations for advancing science and technology policies. It reflects my experience in high-level policy formulation and international collaboration. [STIP Review](#) 

Qualifications and Memberships

Research Member, at Center for Artificial Intelligence and Digital Policy

Since Sept. 2024

President, at The Graduate Public Policy Association

Sept. 2022 to May 2023

Certificate of completion of a **6-month Data Analyst Path program**, Dayche Group, Tehran, Iran

April. 2019 to Oct 2019

Admitted to and successfully completed the **Big Data for Developing Countries** workshop, Held by IKCEST, ISTEP, Xi'an Jiaotong University, and UNESCO, Xi'an, China

September 2018

Participated in the **Future in the Making** conference, Held by The Joint Research Center, EU, Brussels, Belgium

June 2018

Certificate of completion in **Building Technological Capabilities for Sustainable Development through International Collaboration**, UNCTAD, Tehran, Iran

2017

Certificate of completion in **Local Content Policies to Foster Sustainable Local Industrial Development**, UNCTAD, Tehran, Iran

2016

Certificate of completion in **Statistical Tools and Methods for Innovation Data Analysis**, Circle Group at Lund University, Tehran, Iran

2016

Certificate of completion in **Science, Technology, and Innovation Indicators**, UNESCO Institute for Statistics, Tehran, Iran

2015

Certificate of completion in **Design and Evaluation of Innovation Policy**, UNU-MERIT, Tehran, Iran

2015

Skills

Programming: Python, R, SQL (MySQL), STATA, SPSS, RESTful APIs

Research Methods: Mixed-methods research, Survey design and deployment, Usability testing, Focus groups and interviews, A/B testing, Experimental design

Frameworks and Libraries: TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, HuggingFace

Data Analysis and Statistics: Descriptive and inferential statistics, Regression analysis, ANOVA, T-tests, Clustering and segmentation, Factor analysis, Hypothesis testing, Cross-tabulation, Time Series Analysis

Tools and Platforms: Amazon AWS, Qualtrics, Google Forms, Survey Monkey, Figma, Tableau, NVivo, Microsoft Office, Github, Jupyter, VS Code